BAHIR DAR UNIVERSITY institute of technology

Faculty of computing department of software engineering

2nd YEAR

Operating system and system programming

Individual assignment

Windows 11 operating System

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Section A

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1.0 Introduction to Windows 11

Windows 11 refers to the latest version of the operating system from Microsoft, which aims to offer improved user experience, increased security, and better performance than earlier versions. Windows 11 has a revamped look, improved window management capabilities, and enhanced compatibility with contemporary computer hardware and software.

Modern-day computers have become a necessity in all aspects of our lives since we need to interact with machines through the use of an operating system. Windows 11 is among the popular operating systems, mainly due to its stability, efficiency, compatibility, and usability.

The current project entails the installation of Windows 11 on a computer in a virtualized environment. This will offer insight into several concepts like system configuration, file systems, and system-level operations. In addition, there will be the development of a system-level function, thus offering a better understanding of how operating systems operate at the system level.

2.0 Historical Background of Windows 11

A historical background of the creation of Windows 11 should be viewed through the prism of evolution. This product is the result of decades of developing Windows operating system family by Microsoft. Technological changes, customer demands, as well as Microsoft's changing strategies influenced this product's creation. The following historical background of Windows is presented in a step-by-step manner below.

Early History of Microsoft Windows Operating System

This product was launched by Microsoft in 1985 as an upgrade of graphical interface for MS-DOS. Throughout years, such systems as Windows XP, Windows 7, and Windows 8 were created, which served as a perfect base for future products of this kind.

Windows 10 (2015):Windows 10 was another important milestone in the strategy of Microsoft. In fact, "the company has decided to deliver Windows as a service". Therefore, no more upgrades were expected in this case. Windows 10 was expected to become the last one.

Need for Creating a New OS:Some problems were identified by the developers during the usage of the previous one:

An outdated design

System complexity

Security issues

Need for supporting modern hardware and hybrid working conditions

Development of Windows 11

The development of Windows 11 took place under the code name “Sun Valley.” It runs on the same core architecture as Windows 10, although it is improved greatly. Early releases of the operating system were tested in the Windows Insider program in 2021.

Release of Windows 11 (2021)

The release of Windows 11 took place on June 24, 2021, followed by its actual release on October 5, 2021. It was a free update for Windows 10 users who qualify for that privilege.

Major Enhancements

Simpler and modern interface

Multitasking made easier (Snap Layouts)

Improved security (TPM 2.0 required)

Better performance and compatibility

Continuous Development

Further developments of Windows 11 include the introduction of AI-based features, increased system performance, and enhanced security.

3.0 Motivation

The need to develop Windows 11 is determined by the necessity to create an operating system that meets the contemporary requirements associated with high performance, security, and ease of use. Technological progress is rapid and requires operating systems that can accommodate new hardware architectures, cloud computing, and other advances in IT.

▪Need for a Modern OS

Operating systems are created in accordance with specific hardware and user demands of a certain era. However, today users expect from an operating system:

Higher performance

The ability to multitask

Improved graphical user interface (GUI)

Compatibility with cloud services

Windows 11 is aimed at addressing these issues.

▪Enhancing User Productivity

Another motivation to create Windows 11 includes the opportunity to increase productivity thanks to the implementation of:

Snap Layouts and Snap Groups to arrange multiple windows properly

Virtual Desktops to separate spaces depending on the activities

Improvement of the Taskbar and the Start menu to facilitate launching apps

These measures allow minimizing the time needed to switch between tasks.

▪Enhanced Security

The security feature plays a crucial role in the development of the current generation operating systems. The Windows 11 operating system is engineered to have:

A hardware security feature (such as TPM 2.0)

The secure boot system

A protection from various types of malware and cyber security issues

The above features enable increased protection of users' information.

▪Windows 11 compatibility with emerging technology trends

The Windows 11 is engineered to be compatible with the following technological features:

New hardware architecture and processors

Cloud computing

Graphics technology and gaming

4.0 Objectives

The objective of this project is aimed at learning theoretical and practical aspects of Windows 11 through virtualization and system configurations.

✔ Installation of Windows 11 through Virtualization

This objective is aimed at installing Windows 11 through the use of a virtual machine that provides the following benefits:

Isolation from the main computer

Easy to configure and reconfigure

Effective utilization of computing resources

✔ Configurations in the Operating System

In this objective, we study various configurations of the system resources, including:

Configuration of CPU

Configuration of Memory (RAM)

Configuration of Storage

✔ Learning about File Systems

NTFS file system is widely used in Windows 11 for its advanced functionalities. In this objective, we will learn:

Structure of file storage

Permissions and security in the system

Management of large files and directories

We also explore the other file systems like FAT32 and exFAT.

✔ Implementation of System Calls

It is important to understand how system calls are implemented to provide an interface between user-space applications and operating system kernel. Some specific objectives include:

Understand basic system calls and their functioning

Learn some specific system calls such as memory management (locking memory regions)

How hardware is accessed by applications through the OS

✔ Understanding Virtualization

Virtualization is a core concept in modern computing. This objective involves:

Learning how virtual machines operate

Understanding hypervisors

Exploring resource sharing between host and guest systems

5.0 System Requirements

In order to install Windows 11 on a virtual environment, there are certain hardware and software prerequisites that need to be fulfilled.

Hardware Requirements

•64-bit Processor

A processor having 64-bits architecture is a prerequisite since Windows 11 is not compatible with 32-bit processors.

The capacity of processing large amounts of information efficiently.

Allows for the processing of advanced instruction sets by applications.

•At Least 8 GB RAM

Random Access Memory is required for the simultaneous functioning of the host and virtual machines.

8 GB ensures:

Smooth working under multitasking

Proper functioning of the virtual machine

Less RAM would make the machine lag and possibly crash.

•At Least 50 GB Hard Drive Space

Hard drive space is needed for the installation of the following:

Operating system Windows 11

Software applications and updates

Software Requirements

•Oracle VM VirtualBox

Oracle VM VirtualBox is a software for virtualizing machines.

Main Characteristics:

It allows running various operating systems on one device

Supports hardware virtualization technology

Provides customization of system resources

Functions in This Work:

Will be used as an environment for installing Windows 11

Will simulate hardware, like CPU, RAM, etc.

•Windows 11 ISO Image

ISO image is an exact copy of a disk, containing all the necessary files for operating system installation.

Main Characteristics:

Contains all system files

Is used for installing Windows 11 on a virtual machine

Will work as a virtual installation disk

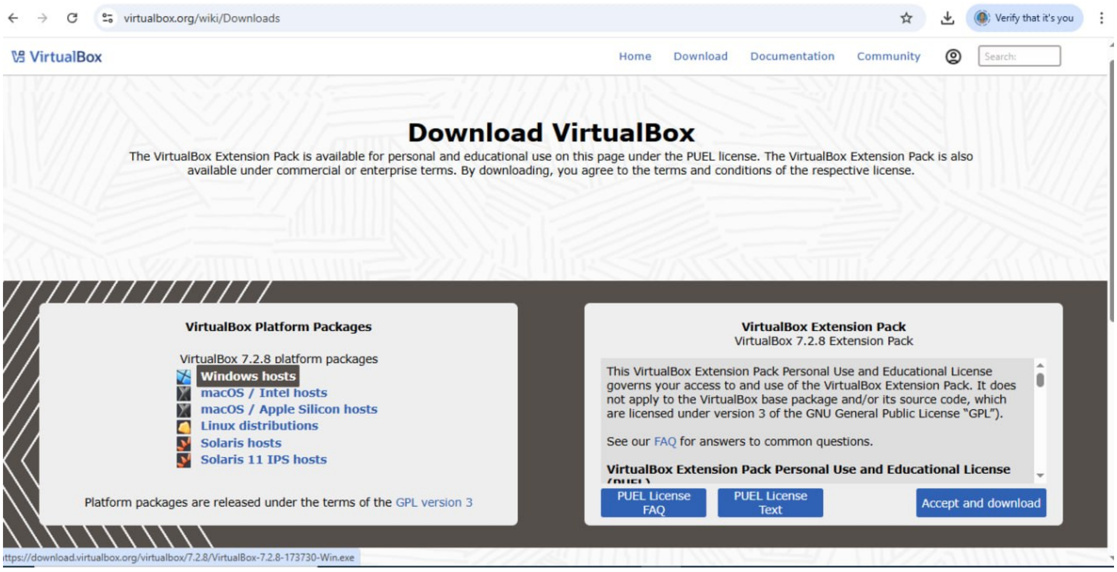
Functions in This Work:

Will enable Windows 11 installation without using any media

6.0 Installation of Windows 11 on Virtual Machine

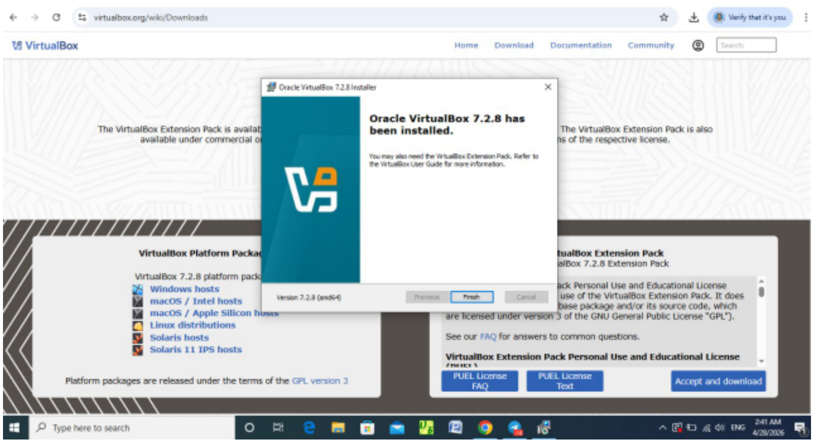
Step 1: Download VirtualBox

Download VirtualBox from its official website to create and run virtual machines.



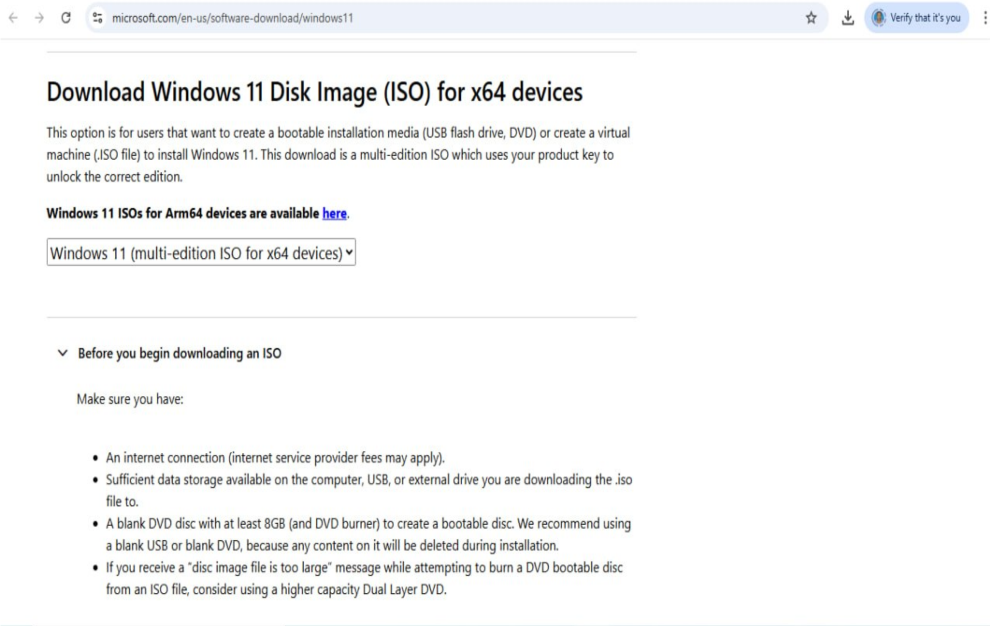
Step 2: Install VirtualBox

Run the installer and follow the setup instructions to install the software.



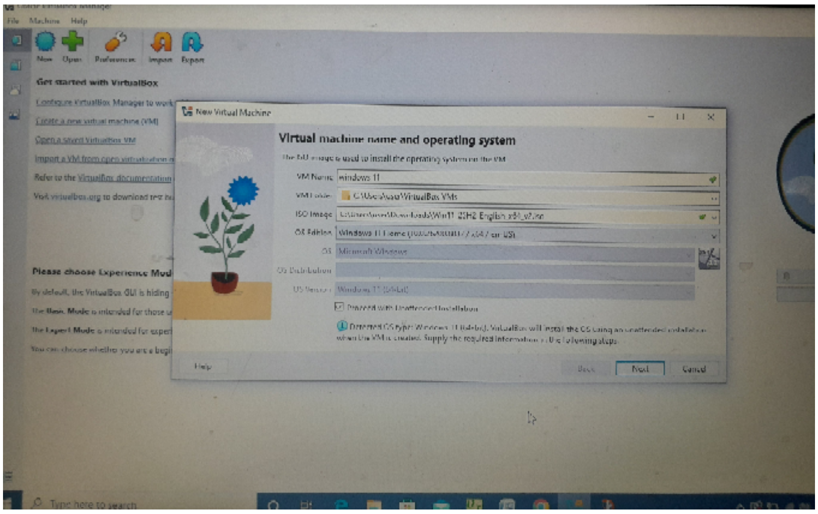
Step 3: Download Windows 11 ISO

Download the Windows 11 ISO file, which will be used to install the operating system.



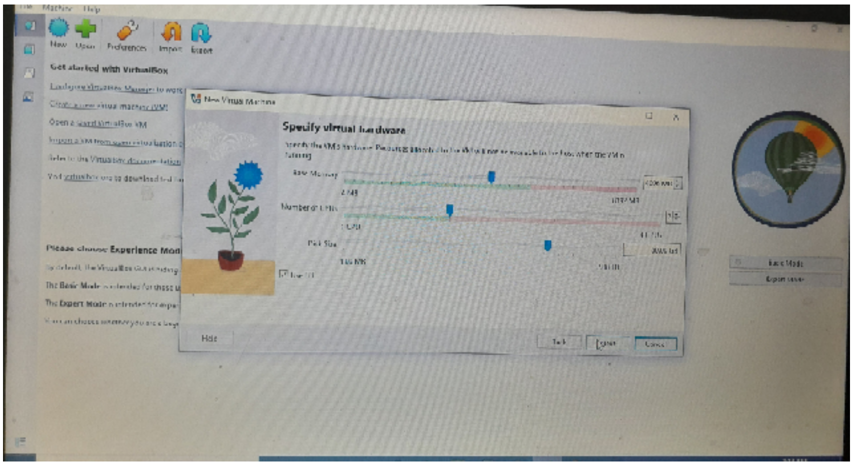
Step 4: Create Virtual Machine

Open VirtualBox, click “New,” and create a virtual machine for Windows 11.



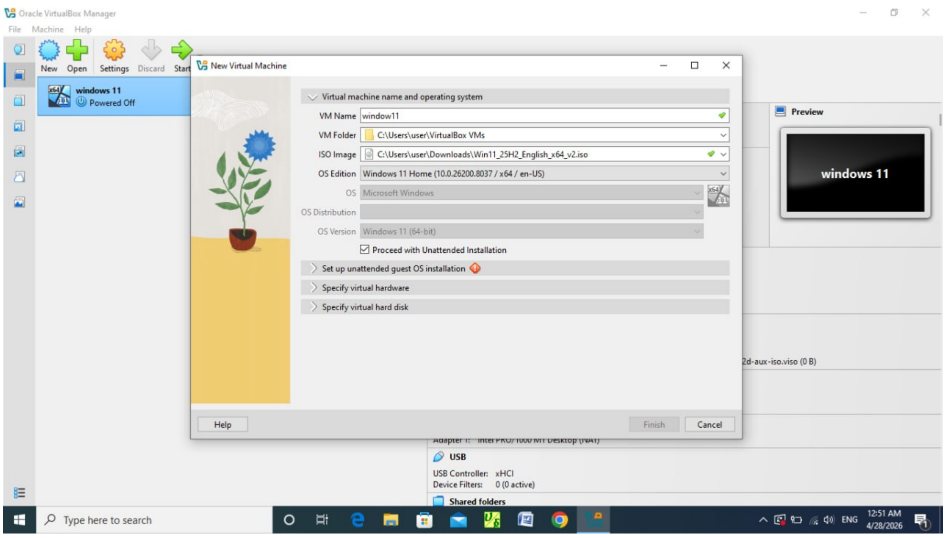
Step 5: Configure RAM and Storage

Assign memory (RAM) and create a virtual hard disk for the virtual machine.



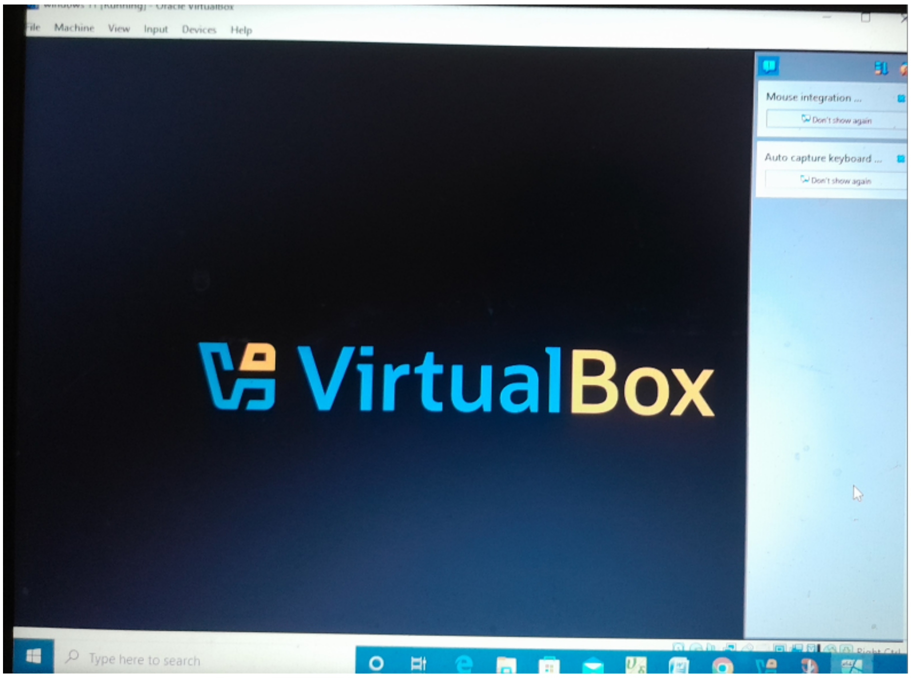
Step 6: Attach ISO File

Load the Windows 11 ISO file into the virtual machine settings.



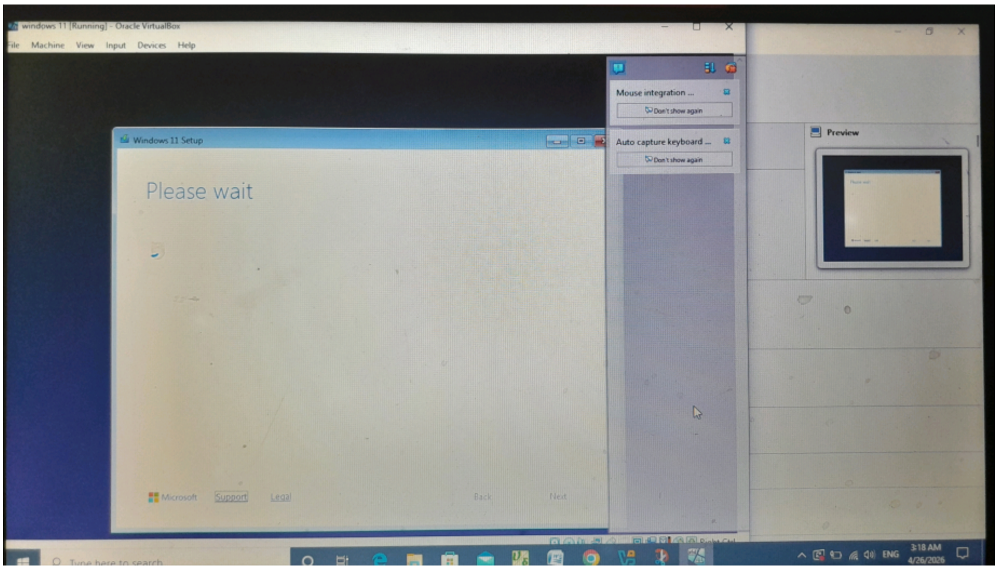
Step 7: Start Virtual Machine

Start the virtual machine to begin the installation process



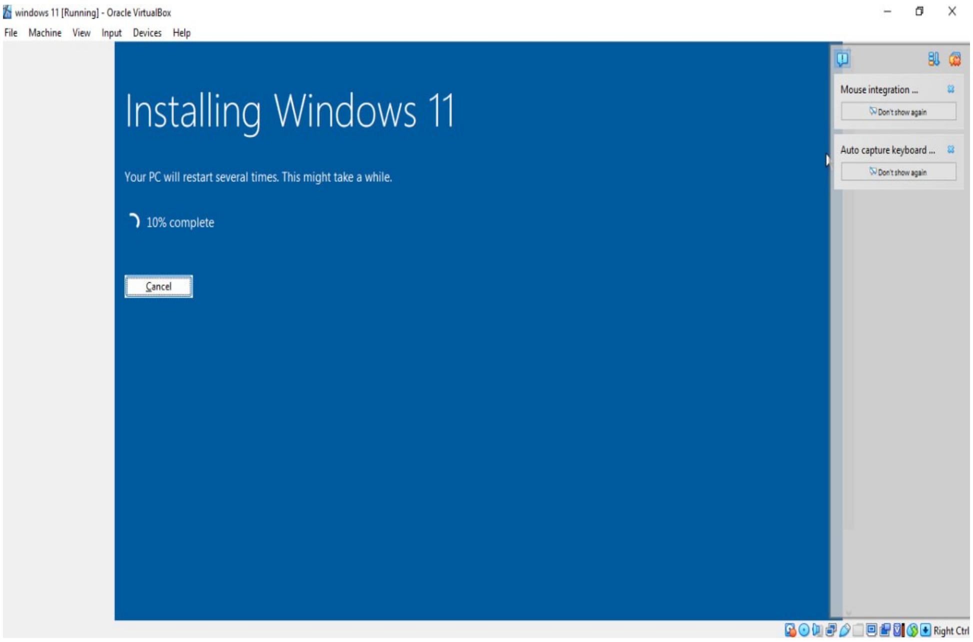
Step 8: Windows Setup Screen

Select language and settings, then click “Install Now.”



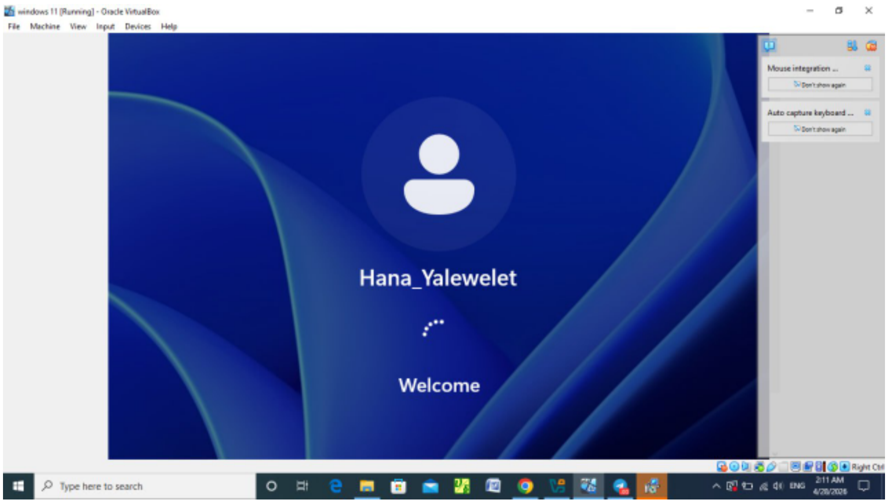
Step 9: Installation Process

Windows installs files and restarts automatically.



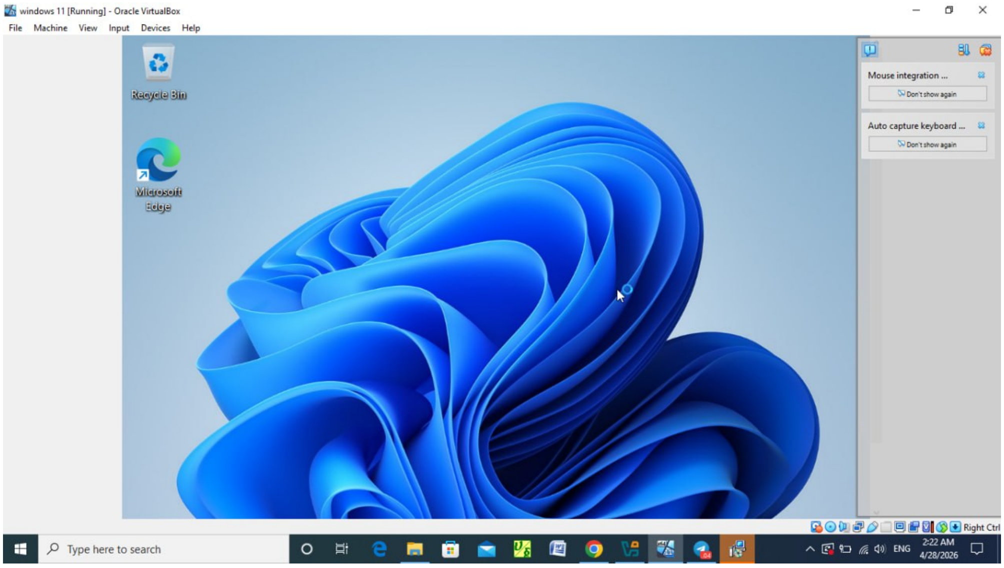
Step 10: Create User Account

Enter username and settings to create your account.



Step 11: Final Desktop

Windows 11 setup completes and shows the desktop.



7.0 Problems Encountered and their Solutions

When it comes to installation, configuration, and working of Windows 11 (particularly in conditions of virtual machines or even when carrying out system level programming such as mlock()), various issues can be encountered. It is important to note them and come up with possible solutions to the same.

Problem and its Solution

✔ Hardware Compatibility Problems

Problem:

One of the main issues with Windows 11 is the need for specific hardware compatibility including:

TPM (Trusted Platform Module 2.0)

Supporting the Secure Boot feature

At least 4GB RAM

Compatible CPU architecture

Some of these specifications might not be available in some machines.

Solutions:

For physical machines, enabling TPM and Secure Boot in BIOS or UEFI.

In virtual machines:

Using virtualization platforms which have support for Virtual TPM

Making installation/configuration changes (only for academic purposes)

.

✔ Virtual Machine Performance Problems

Problem:

Using Windows 11 in a virtual machine can cause problems like:

Poor performance

Interface lagging

Crashes

Possible causes:

Inadequate amount of RAM allocated

Insufficient number of assigned CPU cores

No hardware acceleration

Solutions (steps):

Adjust virtual machine settings.

Allocate more RAM (recommended: 4-8 GB).

Allocate 2 or more CPU cores.

Enable hardware virtualization (Intel VT-x / AMD-V).

Activate 3D acceleration (if supported).

Install guest additions or integration tools.

✔ System Call (mlock()) Problems

Problem:

The system call mlock(), which locks memory into RAM, may fail on Windows-based systems due to the following reasons:

The mlock() is a Linux/Unix system call.

Windows has a different set of system calls/APIs to manage memory.

Solution:

Recognize that Windows uses other functions, such as:

The VirtualLock() Windows API function.

In cross-platform or academic environment:

Use Linux-based virtual machines.

Modify your program accordingly.

✔ Installation Failures

Problem:

"This computer does not meet the requirements for Windows 11"

Installation halted or failed

Solution:

Confirm system compatibility.

Activate TPM and Secure Boot features.

Update BIOS/UEFI firmware.

Utilize genuine ISO file.

Validate disk formatting (GPT necessary for UEFI system).

✔ Storage and Disk Problems

Problem:

Inadequate storage space

Disk partitioning issues

Solution:

Allocate at least 64 GB of storage space.

Employ disk management software for formatting/partitioning.

Transform disk to GPT if necessary.

8.0 File System Support in Windows 11

Windows 11 supports several different file systems that serve their own purposes. NTFS is the default and predominant file system.

●NTFS (New Technology File System):NTFS refers to the default file system employed by Windows 11 on its internal storage drives. It enables advanced functionalities such as enhanced security, reliability, and scalability.

Properties of NTFS File System

a) Security

Ability to specify permissions to files and folders.

Utilizes Access Control Lists (ACL).

Facilitates file encryption using Encrypting File System (EFS).

For instance, a file can be restricted to allow access only to certain specified users.

b) Reliability

Enables journaling through log-based approach.

Recovery from unexpected crashes and power failures.

Details: Prior to any modifications, changes are recorded in logs. In case of a crash, NTFS restores data consistency through these logs.

c) Large Files

Allows extremely large file sizes (up to terabytes).

It is ideal for today's complex uses like video editing software and database management systems.

d) Other Functions

File compression

Disk quotas

Shadow copies

●FAT32 (File Allocation Table 32):An older file system

Very compatible with several types of hardware

Disadvantages:

Maximum file size is 4 GB

Insecure (permissions not allowed)

Uses: Used in USBs and other peripheral devices.

●exFAT (Extended File Allocation Table):Developed for flash media

Allows large files (without the 4 GB limit)

Simple to use

Interoperable with multiple operating systems

Uses: SD cards.

9.0 Advantages and Disadvantages of Windows 11

Windows 11 represents a relatively new operating system created by Microsoft to increase productivity, improve safety, and create an advanced environment for users. This operating system features substantial updates in terms of its appearance, performance, and functionality. Nevertheless, like any other OS, it also has its downsides.

Advantages of Windows 11

●New and Advanced Graphical User Interface

In particular, the operating system is distinguished by an innovative GUI with a centered start menu, rounded corners, and improved animations.

The redesigned layout not only helps users organize the workspace but also allows using the most advanced visualization technologies to boost performance.

●Higher Operating Speed and Performance

One more advantage of the operating system is its improved performance and optimization that allow increasing operating speed and efficiency of multitasking. Namely, the system gives preference to applications that work currently in the foreground mode.

● Advanced Multi-tasking Feature (Snap Layouts)

Another advanced feature introduced by Windows 11 is the Snap Layouts function. It enables users to organize various open windows in particular layouts, facilitating multi-tasking such as reading notes, joining online classes, and programming.

●Enhanced Performance for Gamers

For gamers, Windows 11 has introduced two advanced features which include Auto HDR and DirectStorage. Auto HDR helps enhance the visual effect of the game by ensuring increased color and brightness for games that support Auto HDR. On the other hand, DirectStorage facilitates the fast transfer of data from storage to the graphics processing unit.

●Improved Security

Security has been vastly improved on Windows 11 by introducing the requirement for Secure Boot and Trusted Platform Module (TPM) 2.0. With Secure Boot and TPM 2.0, a computer will be free from any form of attack by malicious software or malware.

●Improved Updates Process

The update process in Windows 11 has also been enhanced such that updates are smaller in size and easier to install compared to previous operating systems. Updates will now run in the background without interrupting user activity.

Disadvantages of Windows 11

●High System Requirements

The new operating system is only able to run on up-to-date hardware with support for TPM 2.0 and rather fresh processors.

Thus, it becomes impossible for some outdated devices to install and use the latest Microsoft OS.

●Incompatibility with Legacy Software and Hardware Components

Some of the older programs might work inappropriately when installed on Windows 11 because of the changes made by the developers.

In such cases, some users will have difficulties installing software that they need for work.

●Limited Opportunities to Customize the System

The new version gives fewer possibilities for customization compared to its predecessors.

This is particularly evident in regard to taskbar and Start menu customization opportunities.

●High Learning Curve for New Users

The changes made in the interface and the change in location of system settings might cause problems for those who have used the previous versions like Windows 10 before.

Some functionalities will not be easily accessible and this will make things difficult and less productive at first for these users.

●Increased System Resources Usage in Low-End Devices

Windows 11 is designed for use in advanced hardware, but it will not work well with lower-specifications hardware due to excessive usage of visual elements and background programs.

●Requirement for Microsoft Account Setup

The installation process of Windows 11, especially the Home editions, necessitates using a Microsoft account during installation.

This poses a risk to user privacy and makes off-line functionality less feasible.

10.0 Virtualization in Windows 11

Virtualization is the creation of a virtual version of a computer system, allowing multiple operating systems to run on one physical machine.

Types of Virtualization

Hardware Virtualization: Uses CPU features (Intel VT-x / AMD-V) to run virtual machines efficiently.

Application Virtualization: Runs applications in an isolated environment.

Desktop Virtualization: Allows multiple desktops on a single system.

3.3 Built-in Tool

Hyper-V: Windows 11's native virtualization platform.

3.4 Setup Steps (Hyper-V)

Check system requirements (64-bit CPU, 4GB+ RAM, virtualization enabled).

Enable virtualization in BIOS (Intel VT-x / AMD-V).

Turn on Hyper-V in Windows Features and restart.

Create a virtual machine using Hyper-V Manager.

Start the VM and install the OS.

3.5 Advantages

Efficient resource use

Safe testing environment

Run multiple OS

3.6 Disadvantages

Requires strong hardware

May reduce performance

Can be complex for beginners

11.0 System Call Implementation

```
#include <windows.h>

#include <iostream>


int main() {

    const int SIZE = 4096; // 4 KB memory

    void* memory = VirtualAlloc(NULL, SIZE, MEM_COMMIT | MEM_RESERVE,
PAGE_READWRITE);


    if (memory == NULL) {
```

```
        std::cout << "Memory allocation failed!" << std::endl;
        return 1;
    }

    // Lock memory (System Call)
    if (VirtualLock(memory, SIZE)) {
        std::cout << "Memory successfully locked in RAM." << std::endl;
    } else {
        std::cout << "Memory lock failed!" << std::endl;
    }

    // Unlock memory
    VirtualUnlock(memory, SIZE);

    // Free memory
    VirtualFree(memory, 0, MEM_RELEASE);

    return 0;
};
```

This program demonstrates a system call in Windows 11 using the VirtualLock() function.

VirtualAlloc() allocates memory.

VirtualLock() locks the memory in RAM so it cannot be moved to disk.

VirtualUnlock() releases the lock.

VirtualFree() frees the allocated memory.

This shows how applications interact with the operating system through system calls for memory management.

Conclusion

The successful deployment of Windows 11 on a virtual machine provides an example of the application of concepts associated with the operating system and the use of virtualization technologies. With the help of available virtualization tools, the operating system could be installed and operated safely without affecting the computer hardware.

During the course of this assignment, important concepts including requirements of the operating system, management of files, as well as basic functionality of the system, have been examined. Also, an implementation of system-level procedure made it possible to learn about memory allocation techniques used by the operating system.

Overall, this task allowed for increasing the knowledge and experience in working with the operating system as well as its components. At the same time, it has once again shown the power of virtualization technology.

Recommendations for Future Development of Windows 11

The operating system in question is highly advanced and effective but requires further refinement. Firstly, it is necessary to lower system requirements to ensure better efficiency on old devices, which would make the operating system available to a larger number of users. Students and people from underdeveloped regions should not be deprived of the chance to utilize such powerful technology.

Software and hardware compatibility are among the key areas requiring improvement since while Windows 11 works well with contemporary software, there still remain problems with legacy applications and hardware. Better compatibility would enable a smooth transition from older versions without losing functionality.

Efficiency is another important characteristic that must be optimized. To do so, it would be sensible to lower the number of background processes and optimize resource management. In addition, security features have to be developed even further with more user-friendly control options to give users the ability to customize their settings to ensure maximum security.

Finally, it would be useful to improve virtualization and other development options to ensure better usability of the operating system by students studying software.

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Microsoft Learn

Microsoft Corporation. Windows System Programming and API Documentation.

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Operating System Concepts

Operating System Concepts

Abraham Silberschatz, Peter B. Galvin, Greg Gagne.

Wiley Publications.